

# BT503 ENVIRONMENT BIOLOGY

**MCQs**

**FOR**

**FINAL TERM**



1: What is not true about bio scrubbers?

Remove toxic metals

2: Following is true about Imhoff tank used in anaerobic filter system?

Can be a single-chamber tank or a two-compartment tank

3: Explosives are characterized by presence of ... group/

Nitro

4: The term “endophytes” is usually used to denote ... microorganisms.

Non-pathogenic

5: UASB stands for?

Up flow anaerobic sludge blanket reactor

6: Facultative ponds which receive raw sewage is also known as?

Primary ponds

7: The bio scrubber is not itself truly a biological treatment system, but rather a ...

Highly efficient method of removing odour components by dissolving them

8: Biodiesel is not derived from?

Algae

9: An aquatic plant?

Macrophyte

10: Important factors to be considered when evaluating and selecting unit operations & processes except?

Sterilization of vessels

11: Dioxins released as by products in a number of chemical processes?

Do not pose threats to environment

12: Non-specific enzyme:

Oxygenase

13: Waste water is applied to the soil, supplying water & nutrients necessary for plant growth. Which is not true?

Generation of aerosols

14: Suitable temperature:

20-30°C

15: What is false about collection stage?

This is the stage when solid waste is transported to the final disposal site

16: False statement about on site-disposal of waste:

Waste volumes are large

17: Which of the following does not use to make vessel of biofilter?

Formic sheet

18: Preliminary treatment involves the removal of?

Coarse solid

19: In bacteria, genetic transfer of materials occurs through?

Plasmid

20: Wolffia globosa is an important tool for showing ... sensitivity.

Cadmium

21: Optimum pH for local environment conditions lies in the range?

6.5-7.5

22: ... uses high temperature for fuse contaminated material.

Vitrification

23: A step in the development of biosensors is?

The immobilization of the biological component at the transducer surface

24: Association of fungi with roots of vascular plants?

Mycorrhizae

25: The ability of a single compound to be degraded can be affected by the presence of other contaminants?

Co-metabolism

26: Which is true for environmental biotechnology?

Enhance or optimize conditions for existing biological systems to make their activities happen faster or more efficiently

27: Thermophiles thrive at temperature above?

45°C

28: ... are compounds which are man-made.

Xenobiotics

29: Natural estrogens are deactivated in humans by?

Glucuronidation

30: Collection of total genomic DNA from an organism?

Genomic libraries

31: ... refers to how pollutants enter a food chain.

Bioaccumulation

32: Which of the following is not a possible destination of the pollutant in the soil?

Appearance in surrounding air

33: The presence of metals in environment may be detected visually by the use of bacteria. One such example is *Vogesella Indigofera* that gives an idea of metal contamination by producing ...

Colored pigments

34: How much arsenic can be accumulated by using *Pteris vittata*?

2.3%

35: Which statement about chlorinated aliphatic hydrocarbons is correct?

Oxidative dehalogenation of CAHs is carried out by facultative aerobes

36: In the conventional activated sludge system, continuous arrival of food in the form of BOD to the aeration tank, leads to rapid and continuous bacterial growth. All of the following problems caused by increased bacterial growth. Except?

The over-all efficiency of the process is enhanced

37: Which kind of the technique are suitable for bioremediation?

In-situ

38: Un-like other marine species, ... are characteristically untroubled by biofouling and study of their skin have established that its unique texture coupled with its inherent antimicrobial properties help keep them clean of fouling organisms.

Sharks

39: Which plant is used as a model to protect from various agents?

Tobacco plant

40: The airflow in the submerged aerated biofilter is always ... while the liquid flow can be ...

Up-flow, downflow

41: Which of the following approaches do not remove the pollutants but only immobilize the pollutants?

Phyto-stabilization

42: The ... product of pollutant may be more harmful than the original pollutant itself, therefore, it is important to understand the chemistry of the pollutant.

Breakdown

43: Among the first synthetic organic chemicals to create environmental problems were ...

Detergents

44: Immobilization refer to the removal of contaminants, typically metals, by means of ... by various microbes or plants species.

Adsorption

45: Which of the following factors has no effect on the execution & implementation of environmental technology?

Health of the general population

46: Dioxins have many congeners. The most toxic is ... which is thought to be the most poisonous synthetic chemical.

TCDD (Tetra-chloro-dibenzo-p-dioxin)

47: Which of the following methods are best suited to instances of relatively localized pollution within a site, typically in “hot-spots” of

medium to relatively high concentration which are fairly near to the surface.

### Ex-situ

48: For land-based applications, the soil type best suited biotechnological inventions are ...

### Sand & gravel

49: Plants containing the novel DNA grow, a gene coding is frequently for ...

### All

50: Deliberate transfer of a gene from one organism to another is called:

### Foreign gene

51: Western blotting technique is used for the detection of ...

### Protein

52: Which of the following have the innate ability to degrade hydrocarbons contaminants through metabolic action and biosurfactant of hydrocarbons?

### HCB (Hydrocarbonoclastic bacteria)

53: What is false about poor biodegradation of hydrocarbons?

### Easily degraded by enzymatic attack

54: Liquid fuels are of considerable importance due to ...

The relative ease of transport & handling and inherent controllability of combustion

55: Which process used in the natural process of dewatering?

### Drying beds, sludge lagoons

56: In aerobic co-metabolism of CAHs ... is required.

Successful co-metabolism can be achieved by ensuring that primary substrate is well in excess of that needed to overcome toxicity from the CAHs or its transformation products

57: Insect virus that causes over expression of genes in molecular biology.

### Baculovirus

58: Acute toxins

### All

59: By using recombinant technologies, scientists are now producing ...

### Insulin

60: ... are combustible materials, containing within themselves all oxygen needful for their combustion, which burn but do not explode, and function by producing gas which produces an explosion.

### Propellants & low explosives

61: ... are type of waste stabilization pond used for biological treatment of industrial and domestic waste water.

### Facultative lagoons

62: Pentane is utilized as a ... in the preparation of phenols.

### Pentane

63: Polystyrene, as a propellant for aerosol spray, as a filling of low-temperature ... and as reference compound in gas chromatography.

### Thermometers

64: Reintroduction of these ... bacteria to the polluted sites should give them an advantage over the indigenous bacteria as they would be better suited to survive and remediate the contamination.

Trained

65: The DNA construct contains ... which are complementary to the plant DNA to enable the inserted piece to recombine into the plant genome.

Regions of DNA

66: Attack by insects not only causes damage to the plant but also provides a route ...

For bacterial or fungal infection

67: Bacteria which thrive at temperature above 85°C is called as ...

Hyper-thermophiles

68: Which attribute of pollutant is not linked to biomagnification?

Long-lived, mobile, soluble in fats, biologically active

(Other than these options)

69: Following substances cannot use to make the vessel of biofilter?

Concrete, metal or durable plastic (Other than these options)

70: In general, enzymatic biosensors are based on ... by different classes of compounds.

The selective inhibition of specific enzymes

71: Which enzymes are used in leather industry?

Pancreatic

72: Toxicity represents the potential damage to life and can be both short and long term. It is related to the:

Type of the pollutant, concentration & the time of exposure

73: Bio-filters can reduce odour release by ...

95% or more

74: For higher efficiencies of phosphorus removal, effluent polishing methods can adopt ...

Addition of coagulants

75: In Imhoff tank, settling occurs in the ...

Upper compartments (settling compartments)

76: In the soil, a pollutant has basically ... possible destinations.

4

77: Biodiesel is derived mostly from ...

Vegetable oil

78: DDT was of great benefit to humans in their fight against ...

Malaria

79: Two characteristics of chlorinated pesticides that led to great harm are:

Their resistance to biodegradation & their hydrophobicity

80: Water treatment accounts for ... of the total global environmental market.

25%

81: Basic equipment required in preliminary treatment involves screen, ...

Grit chamber & flow measurement

82: What is false about final disposal of sludge in ocean?

It has beneficial uses

83: ... refers to the biodegradation of contaminants in the soil by adaphic microbes enhanced by the inherent character of the rhizosphere itself.

Rhizo-degradation

84: A tumor like growth, seen as a crown gall in plants?

*Agrobacterium tumefaciens*

85: Explosives use ... functional group.

Nitro

86: Aerobic co-metabolism of ... requires NADH to provide the 2H.

CAHs-II

87: Non-specific enzymes are ... that initiate the oxidation of hydrocarbons such as methane or toluene.

Oxygenase

88: The metals release in air by all of the following except?

Earthquake

89: *mer* A gene codes for the enzyme ...

Mercuric ion reductase

90: What is true regarding zooplanktons?

Rely on tides & currents as a transport

91: One such recombinant is a plant where the fatty acid composition in the seed has been modified to produce triacylglycerols containing elevated levels of trierucinic acid suitable for use in the polymer industry.

*Arabidopsis thaliana*

92: Factors related to biomagnification except?

Source

93: ... is the branch of biotechnology that addresses environmental problems, such as the removal of pollution, renewable energy generation or biomass production. By exploiting biological processes.

Environmental biotechnology

94: The goals of environmental biotechnology can be achieved in ... ways.

2

95: ... key points for environmental biotechnology.

3

96: The market will have grown to ... US dollars by 2025

7400 billion

97: In general, the use of biotechnology for environmental management relies on ...

Mesophilic microorganisms

98: Extremophiles could provide a way of developing alternative routes to many ...

Conventional chemicals or materials

99: *Thermus aquaticus* means:

Bacterium that can tolerate high temperature

100: Taq Polymerase is a:

DNA polymerase of *Thermus aquaticus*

101: Pfu polymerase is a DNA polymerase cloned from:

*Pyrococcus furiosus*

102: ... survive intensely saline environments, such as exist in natural salt lakes or salt evaporation ponds.

**Halophiles**

103: Acidophiles thrive in the conditions of ... typically below ...

**Low pH, 5**

104: Extremozymes capable of functioning below ... have been isolated.

**pH 1**

105: ... flourish in environments at pH 9 or more such as soda lakes & heavily alkaline.

**Alkaliphiles**

106: Greek word “xenos” meaning:

**Foreign**

107: Which statement about chlorinated hydrocarbons is correct?

**Fixation of carbon by phytoplankton**

108: In planning studies for the implementation of the waste water treatment, the following points must be clearly addressed:

**Environmental impact studies on the receiving body, treatment objectives, treatment level & removal efficiencies**

109: The correct sequence of Initial Steps in Solid Waste Management:

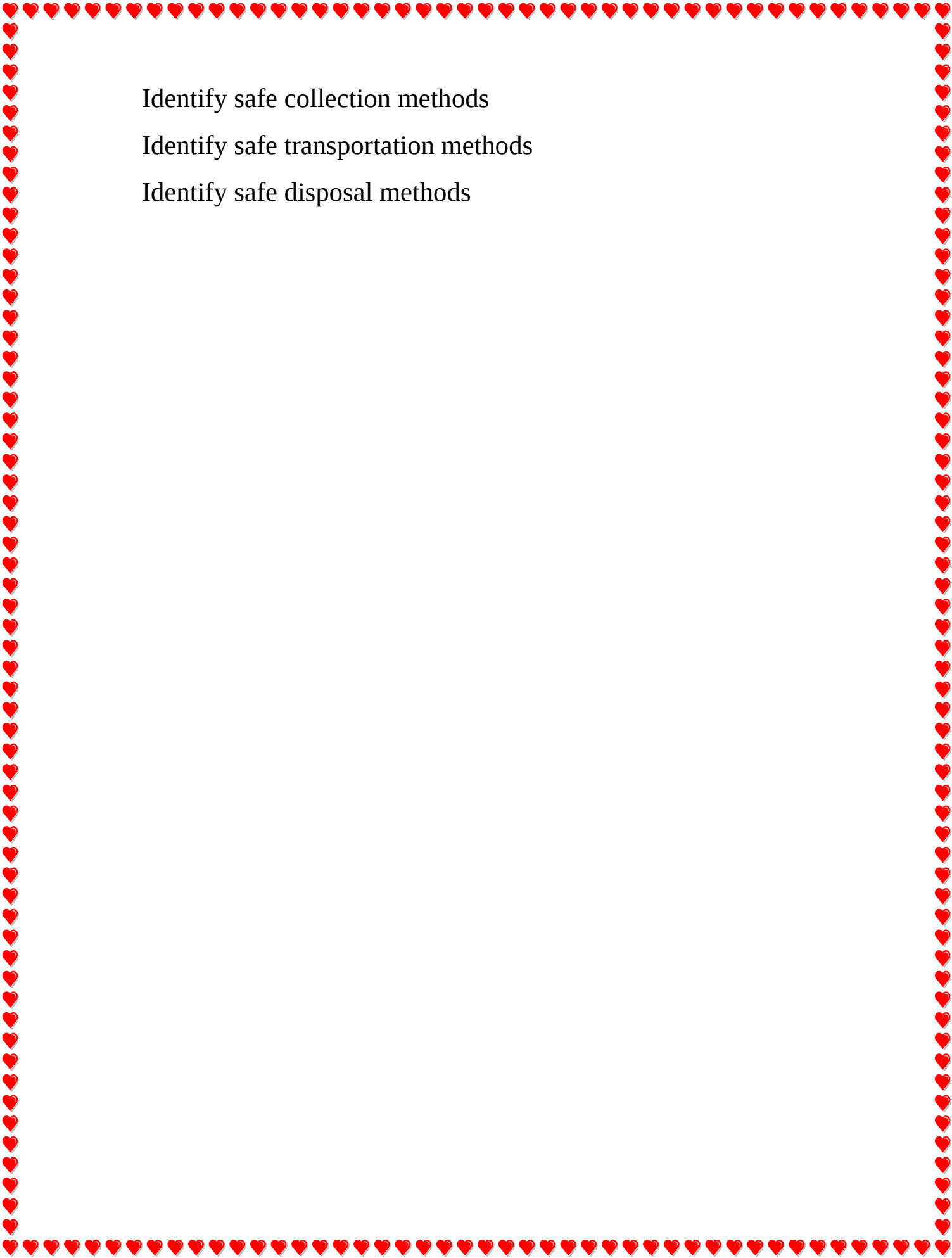
**All**

Identify the types of waste

Identify the source of waste

Determine the potential health hazard from waste

Determine the volume of waste generated



Identify safe collection methods

Identify safe transportation methods

Identify safe disposal methods